

HSE Intelligence — Mobile Architecture · All Roles, All Nine Workflows, Mapped to the AI-ISMS Data Model

Target state. Purple tags show which AI-ISMS data class each screen feeds — the mobile app is the platform's primary source of C-layer (Customer Dynamic) data.

Rev 4.0 · 30 Jul 2026 · workflow numbering per ALGO-ISMS-WF-2026-v1.0

Data classes per AI-ISMS Data Input Architecture Ver 1.0

5 BUILT 4 TO BUILD C-LAYER PRODUCER

LAYER 1 · MOBILE SHELL — shared by all four role apps; capture modalities align to the AI-ISMS multimodal input spec (text, image, video, audio)

Auth & JWT login, refresh, RBAC route	API Client axios + retry	GPS Capture lat/long on all records	Camera photo evidence → C5	QR / Barcode badge · asset · vehicle	Offline Queue draft & sync	Push & Alerts 60/30/7 · check-in miss	Gate Client renders block / amber / override	Audio Capture toolbox & briefing → C6
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WORKER
Employee · Field Operative · Driver-Operator

BUILT

Incident / Near Miss / Unsafe Act

Severity, injury detail, GPS, photos

C3 C5

Report Risk / Hazard

L × S scoring, existing controls

C3

Request Permit

Type, description, station, validity

C1

Checklist & Tasks · Shift check-in

Assigned tasks, start/stop timestamps

C2 C7

TO BUILD — FILLS THE GAPS

My Competence Card

Live matrix vs role. Expiry at **60 / 30 / 7 days**. Shows **which tasks are blocked**. New-worker flag for first 30 days.

B4

Training & Assessment

Course completion, score. **Below pass = flag stays**. Toolbox knowledge feeds the 80% attendance rule.

B4 C6

Fatigue Declaration

Shift hours, consecutive days, night shifts → live **F index** shown before permit request. Non-medical proxies only.

C7

Journey Risk & JMP

Route × Mode × Cargo. **Weather go/no-go** ack. Pre-trip gate. Vehicle QR.

C1 C2

Journey Check-in

Timed — **road every 2 h**; miss alerts control room. Post-movement review: defects, deviations.

C2

My Safety Score

Personal **Human Readiness** contribution — competence gaps, fatigue flags.

C7

SUPERVISOR
Line Manager · Area & Authorising Authority

BUILT

Investigate & 5-Whys

Acknowledge, RCA, days away, signature

C3

CAPA Assignment

Owner, due date, escalation

C4

Permit Acknowledge

Notes, actual times, deviation flag

C1 C2

Safety Walk · Team & Shift

Ratings, attendance, toolbox talks

C3 C6

TO BUILD — FILLS THE GAPS

Team Competence Matrix

Nightly gap report. Expiring certs. **Assign buddy** for new workers on WAH / CS / hot work.

B4

Gate Override Console

F 10–14 acknowledge · **F 15–19** sign-off with mandatory note. Every override captures **reason, context, outcome**.

D4 — CORE FEATURE

RAMS Scoring

6-criterion rubric, 0–20 each. **<60 reject · 60–79 conditional · ≥80 approve**.

C1 B2

Contractor Site Control

Verify induction & certs, site-access status, contractor toolbox completion.

B4 C1

Transport Authorisation

Approve movements scoring **≥ 13**. Set check-in interval and comms protocol.

C1

Check-in Monitor

Live journey board, missed check-in escalation, defect → maintenance action.

C2 C4

Team SPS View

Five domain sub-scores for own area; red-line KPI alerts.

C1

SAFETY MANAGER
HSE Manager · Site Owner

BUILT

Incident Close-out

Approve investigation, lessons, closure

C3

Permit Approvals

Approve / reject, active monitoring

C1

CAPA Sign-off

Validate closure evidence

C4

Hazard Register Review

Open → under review → controlled

B3

TO BUILD — FILLS THE GAPS

SPS Dashboard

Weekly 0–100, **five domain breakdown**. Bands Critical ≥75 · High 50–74 · Elevated 25–49 · Acceptable 10–24 · Low <10.

C1-C4

SPS Alerts & CAPA Lookup

Fires on **Δ ≥ 10 pts / week**, band change, or KPI red-line. 2–3 pre-defined CAPAs per KPI.

C4

Competence Matrix Owner

Role → course → validity. Monthly **training effectiveness**: incident rate trained vs untrained.

B4

Fatigue Exception Authority

Only role able to authorise **F ≥ 20**, with written rationale.

D4

Contractor Pre-Qualification

Insurance,SSIP/CHAS, 3-yr LTIFR/TRIFR. **>2× IOGP = reject; 1.5–2× conditional**.

B1 D1

Contractor Scorecard

Quarterly. **<50 enhanced oversight · <30 contract review**. Two quarters <30 = off list.

C1

Transport Oversight

Weather limits by mode, fatigue flag rate, defect rate, monthly KPI batch.

C2 C7

Change & Drift Log

Procedure updates, equipment mods, staffing changes, temporary arrangements — **MOC-Lite** risk-spike input.

C8

AUDITOR
Safety Officer · Independent Assurance

BUILT

Assigned Audits

Scheduled / in-progress / overdue

C3

Checklist Execution

Pass / fail / N-A, evidence, score

C3 D6

Permit Verification

Valid / not displayed / invalid

C1

Hazard Verification

Confirm hazard actively managed

C3

TO BUILD — FILLS THE GAPS

Competence Evidence Audit

Sample workers on site, verify certs against matrix, flag **any expired safety-critical cert**.

B4

Gate Override Audit

Review every fatigue and competence override — who, when, rationale. The **regulatory defensibility record**.

D4

Data Integrity & Validation

Verify feeds current — any source **>14 days stale = Data Gap**, applying a 10-point SPS penalty. Confidence-score review.

VALIDATION

Contractor Audit

Independently re-score RAMS against the same rubric; on-site compliance findings.

C1

Transport & Vehicle Audit

Pre-trip sampling, licence & roadworthiness, **check-in log completeness**, post-movement reviews.

C2

Checklist Effectiveness Feedback

Flags low-signal checklist items and items that consistently drive findings — input to continuous checklist optimisation.

D6

THE INTEGRATION SPINE · DETERMINISTIC GATE ENGINE — server-side at permit issuance and journey departure. Rule-based and auditable; the AI layer may add context but never changes a verdict.

1 · RAMS Linked Task has a current, approved risk assessment. FROM WF-01	2 · Competence Verified Every named worker holds valid certs for the permit type. Expired safety-critical cert = hard block. FROM WF-06 — NEW · B4	3 · Fatigue Index <10 pass · 10–14 amber ack · 15–19 supervisor sign-off · ≥20 hard block FROM WF-06 — NEW · C7	4 · Zone / SIMOPS No two high-energy permits active within 30 m FROM WF-02 · C1	5 · Contractor Approved Company pre-qualified and RAMS scored ≥ 60 FROM WF-08 — NEW	6 · Weather & Journey Mode operating limits not exceeded; score ≥13 requires Transport Authorisation FROM WF-09 — NEW
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WHAT MOBILE RETURNS TO THE AI-ISMS — the field app is not only a data source; it closes the learning loop the specification calls its Core Feature

C — CUSTOMER DYNAMIC Operational Truth C1 permits · C2 execution reality · C3 incidents & observations · C4 CAPA · C5 photo evidence · C6 toolbox audio & attendance · C7 fatigue proxies · C8 change signals. Mobile produces nearly all of it.	D4 — HUMAN-IN-THE-LOOP Every Override Is Training Data When a supervisor accepts, modifies or rejects a gate outcome, the system records reason, context and outcome — used to cut false positives and build frontline trust.	D6 — CHECKLIST OPTIMISATION Audit Feedback Checklist items consistently flagged as critical drivers are promoted; low-signal items removed. Auditor execution data drives the re-ordering.	VALIDATION STAGE Data Quality Gate Ingestion → Normalisation → Validation → Analysis → Decision → Record → Learning. Stale feeds (>14 days) are penalised, not silently trusted.
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DETERMINISTIC FORMULAE — fixed, human-defined, auditable. Every output traces to an explicit rule. Verbatim from the algorithmic specification.

Fatigue Index — WF-06 · feeds C7 $F = ((\text{Shift Hrs} - 8) \times 1.5) + ((\text{Consec. Days} - 5) \times 2) + (\text{Night Shifts in } 7d \times 3)$ <10 acceptable · 10–14 amber, supervisor acknowledges · 15–19 sign-off before high-risk permit · ≥20 hard block, 8 h rest, Safety Manager exception only	Safety Performance Score — WF-07 $\text{SPS} = 0.25 \cdot \text{Hazard Exposure} + 0.25 \cdot \text{Control Integrity} + 0.20 \cdot \text{Work Discipline} + 0.20 \cdot \text{Human Readiness} + 0.10 \cdot \text{Org. Health}$ Weekly batch. Same five domains as PIRS, but measures current state — it does not predict . PIRS is the AI counterpart.	Journey Risk — WF-09 $\text{JRS} = \text{Route}(1-3) \times \text{Mode}(1-4) \times \text{Cargo}(1-3)$ 1–4 low · 5–12 medium · ≥13 high, requires Transport Authorisation. Check-in: road every 2 h; marine per voyage plan; air per flight plan.	Contractor Gates — WF-08 $\text{RAMS} = \sum 6 \text{ criteria} \times 0-20$ LTIFR vs IOGP benchmark RAMS <60 reject · 60–79 conditional · ≥80 approve. LTIFR >2× benchmark = rejected; 1.5–2× conditional with enhanced monitoring.
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THE NINE WORKFLOWS — algorithmic-spec numbering (as used in the review); master-data reference beneath each

WF-01 Risk Assessment & Hazard ID BUILT MD WF-01	WF-02 Permit-to-Work & CoW BUILT MD WF-02	WF-03 Incident Reporting BUILT MD WF-03	WF-04 CAPA BUILT MD WF-04	WF-05 Audit, Inspection & Compliance BUILT MD WF-05 + 06	WF-06 Training, Competence & Human Readiness TO BUILD MD WF-07	WF-07 Safety Performance Scoring TO BUILD MD WF-08	WF-08 Contractor & High-Risk Work TO BUILD MD WF-09	WF-09 Transport & Logistics TO BUILD MD WF-10
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MOBILE API SURFACE — FastAPI /api/v1 · JWT + org-scoped RBAC · blue endpoints are new

Incidents /incident-workflow/* /near-miss-workflow/*	Permits /permit-workflow/*	Risk · CAPA · Audit /risk-workflow/* /capa-actions/* /audits/*	Competence /competence-matrix/* /training-records/*	Fatigue & Gates /fatigue/index /gates/permit-check /gates/log	Contractor /contractors/* /rams-scores/*	Transport & SPS /journey-plans/* /vehicles/* /sps/score /sps/alerts
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DATA ADDITIONS — aligned to the AI-ISMS core structured data model; every record carries the mandatory cross-cutting metadata

NEW MASTER DATA Vehicle Contractor Company Contractor Worker Training Programme Certification Type Competence Profile Competence Matrix Weather Limit Table CAPA Lookup IOGP Benchmark	NEW OPERATIONAL DATA — incl. AI-ISMS entities not yet present Training Record Competence Gap Fatigue Index Gate Decision Log Override Log RAMS Score Contractor Scorecard Journey Plan Check-in Event SPS Snapshot Work Execution Event Supervisor Interaction Change Event AI Decision Log	MANDATORY METADATA — ALL ENTITIES Created date Last reviewed Last verified Source system Jurisdiction Confidence score AI vs human flag Override history
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● Built & integrated ● To build — fills the gap ● Deterministic gate ● AI-ISMS data class fed

Build order follows the dependency: WF-06 competence and fatigue feed the WF-02 permit gate, so it comes first; WF-08 and WF-09 both consume that gate; WF-07 aggregates all of them and lands last. Fatigue uses non-medical, privacy-safe proxies only — shift length, consecutive days, night shifts, task intensity.